Critical quantum metrology with the quantum Rabi model: Fisher information across the superradiant transition

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Critical quantum metrology exploits the diverging susceptibility of systems near quantum phase transitions to surpass classical estimation limits. We show that the quantum Rabi model, a single qubit coupled to a single bosonic mode, provides such a resource in the classical oscillator limit, where the ratio of qubit to mode frequency plays the role of the thermodynamic limit. Using first- and second-order Schrieffer-Wolff effective Hamiltonians, we derive the quantum Fisher information (QFI) for the estimation of the coupling strength and show that it diverges with a critical exponent of -2 in the normal phase. In the superradiant phase, where the effective potential inverts and standard methods fail, we introduce a displaced and squeezed Fock-basis scheme that restores numerical convergence, and we obtain the QFI of the physical ground state by accounting for the coupling dependence of the basis transformation. The entanglement entropy peaks at the critical point, providing an independent signature of the transition.

I. INTRODUCTION

Quantum metrology aims to exploit genuinely quantum features to estimate parameters with a precision beyond classical limits, with applications ranging from magnetometry to interferometry [1]. Whereas uncorrelated probes are limited to the shot-noise scaling $(\Delta\theta)^2 \sim 1/N$, entangled probe states can reach the Heisenberg scaling $(\Delta\theta)^2 \sim 1/N^2$ [2–4]. One of the most promising strategies relies on systems close to instability [5], in particular at quantum phase transitions (QPTs). These transitions occur at zero temperature and are driven by quantum fluctuations that reorganize the ground state (GS) macroscopically as the spectral gap closes [6], making the system extremely susceptible to small changes in the driving parameters. A drawback is that general QPTs display true critical behavior only in the thermodynamic limit, which severely restricts the available numerical methods.

Superradiant phase transitions (SPTs) [7], which stem from Dicke's concept of superradiance [8], are a prominent class of QPTs, in which a system shifts from a vacuum normal phase to a phase with macroscopic photon occupation. It has been realized that strict criticality can also emerge in finite-size systems such as the quantum Rabi model (QRM) [9], a single two-level atom interacting with a single cavity mode. There, criticality is recovered in the so-called classical oscillator limit, in which the atomic transition frequency becomes much larger than the cavity frequency. This frequency ratio acts as an effective thermodynamic limit.

Following the framework of Garbe et al. [5], we use the quantum Fisher information (QFI) and the Cramér-Rao bound (CRB) to study the estimation of the coupling strength of the QRM near its critical point. We derive effective Hamiltonians by first- and second-order Schrieffer-Wolff (SW) transformations, obtain the analytic QFI in the normal phase, and complement the results with numerical simulations, performed with QuTiP [10], that extend them to the superradiant phase. Critical slowing down [5] is a dynamical consequence of criticality that we do not address: this work focuses on static equilibrium properties. The code used for the numerical analysis and the figures is publicly available [11]. We use natural units, $\hbar = 1$.

II. CRITICAL QUANTUM METROLOGY

A. QFI and the Cramér-Rao bound

Given a family of states ϱ_θ and a POVM $\{\Pi_x\}$, the outcome probabilities are $f(x;\theta) = \mathrm{Tr}(\varrho_\theta \Pi_x)$, and the variance of any unbiased estimator of θ obeys the CRB $\mathrm{Var}(\hat{\theta}) \geq 1/[\nu F(\theta)]$, where ν is the number of repetitions and

$$F(\theta) = \int \frac{1}{f(x;\theta)} \left(\frac{\partial f(x;\theta)}{\partial \theta} \right)^2 \mathrm{d}x \tag{1}$$

is the classical Fisher information [1, 12]. Maximizing F over all POVMs defines the QFI, $F_Q(\varrho_\theta) = \mathrm{Tr}\left[\varrho_\theta L_\theta^2\right]$, where the symmetric logarithmic derivative L_θ satisfies $(L_\theta \varrho_\theta + \varrho_\theta L_\theta)/2 = \partial_\theta \varrho_\theta$. Then $(\Delta\theta)^2 \geq 1/F_Q(\varrho_\theta)$ (we take $\nu = 1$) [13]. For a pure state $|\psi_\theta\rangle$, such as the GS of a Hamiltonian, the QFI reads [5, 13, 14]

$$F_Q = 4\left[\langle \partial_\theta \psi | \partial_\theta \psi \rangle - \langle \partial_\theta \psi | \psi \rangle \langle \psi | \partial_\theta \psi \rangle\right]. \tag{2}$$

B. Criticality in finite-size systems

Consider $H = H_1 + gH_2$ with $[H_1, H_2] \neq 0$ and different GSs for H_1 and H_2 [15]. At the quantum critical point (QCP) $g = g_c$ the spectral gap closes, and the correlation

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length diverges as $\xi \propto |t|^{-\nu}$ with $t = |g - g_c|/g_c$ [16]. A perturbation of a localized constituent therefore spreads to the whole system, which is the origin of the enhanced susceptibility that critical metrology exploits. Strictly, a sharp transition requires the thermodynamic limit, since in finite systems the gap only has an avoided crossing. The QRM is nevertheless a finite system with a true transition, as can be seen from its multi-spin generalization, the Dicke model [7],

$$H = \omega a^\dagger a + \sum_{i=1}^{N} \frac{g}{\sqrt{N}} \sigma_{ix}(a + a^\dagger) + \sum_{i=1}^{N} \Delta \sigma_{iz}. \tag{3}$$

Introducing a generic scaling parameter C, so that the order parameter is $\langle a^\dagger a\rangle/C$, the couplings $\Omega = \omega C/N\Delta$ and $\gamma = g\sqrt{C}/\Delta\sqrt{N}$ must stay finite as $C \to \infty$. This can be achieved with the usual thermodynamic limit $N \to \infty$, or with the *classical oscillator limit* $\Delta/\omega \to \infty$, $g/\omega \to \infty$ at fixed γ and Ω [7]. For the QRM ($N = 1$) the order parameter scales as $\langle a^\dagger a\rangle/(\Delta/\omega)$, so the ratio Δ/ω plays the role of the system size.

III. THE QUANTUM RABI MODEL

The QRM describes a two-level atom of level spacing 2Δ interacting through the dipole coupling with a single quantized cavity mode [17],

$$H_{\text{Rabi}} = \omega a^\dagger a + \Delta\sigma_z + g\sigma_x(a + a^\dagger), \tag{4}$$

where a and $a^\dagger$ are bosonic operators, ω is the mode frequency and g is the coupling strength (with dimensions of frequency). The QRM was solved exactly by Braak [18], but this solution is not suited to obtaining closed-form metrological bounds, so we follow the standard approximate approach of critical metrology [5, 13]. It is customary to use the dimensionless coupling $\gamma = g/\sqrt{\Delta\omega}$. We will estimate g from the GS family ϱ_g. The same analysis applies to other parameters, such as Δ [5].

A bosonic mean-field treatment, replacing $a \to \langle a\rangle$, reduces H_{Rabi} to a two-level problem whose QFI is $A^2\Delta^2/(\Delta^2 + A^2g^2)^2$, with $A = 2\,\text{Re}\langle a\rangle$. It does not diverge, so it fails to capture criticality, and the model must be treated beyond mean-field.

IV. ANALYTICAL RESULTS: SCHRIEFFER-WOLFF EFFECTIVE MODELS

The SW transformation is a unitary change of frame, $H_{\text{eff}} = e^S H e^{-S}$, that decouples subspaces and yields a low-energy effective Hamiltonian [19]. We partition $H = H_0 + H_1$ with the even part $H_0 = \omega a^\dagger a + \Delta\sigma_z$ and the odd perturbation $H_1 = g\sigma_x(a + a^\dagger)$. The general construction and the conditions on the generators are given in Appendix A.

A. First-order transformation (SW1)

The condition $[s_1, H_0] = -H_1$ (Eq. (A3)) is solved by $s_1 = \frac{2ig\Delta}{4\Delta^2-\omega^2}\sigma_y(a + a^\dagger) + \frac{g\omega}{4\Delta^2-\omega^2}\sigma_x(a - a^\dagger)$, which in the classical oscillator limit $\omega/\Delta \to 0$ reduces to

$$s_1 = \frac{ig}{2\Delta}(a + a^\dagger)\,\sigma_y. \tag{5}$$

Using $[s_1, H_1] = \frac{g^2}{\Delta}(a + a^\dagger)^2\sigma_z$, Eq. (A4) gives

$$H'_{\text{SW1}} = \omega a^\dagger a + \Delta\sigma_z + \frac{g^2}{2\Delta}(a + a^\dagger)^2\sigma_z. \tag{6}$$

Critical point. An exact unitary transformation preserves the location of the QCP [20]. Projecting onto the lower spin eigenspace and using a bosonic mean-field amplitude α, the energy as a function of $|\alpha|^2$ is

$$E_I(|\alpha|^2) = -\Delta + \left(\omega - \frac{2g^2}{\Delta}\right)|\alpha|^2, \tag{7}$$

whose slope changes sign at

$$g_c = \sqrt{\frac{\Delta\omega}{2}}, \qquad \gamma_c^2 = \frac{1}{2}, \tag{8}$$

consistent with Ref. [5]. For $g < g_c$ the vacuum is a stable minimum; for $g > g_c$ the energy is unbounded below, signaling that SW1 breaks down in the superradiant phase.

Ground state. Projecting onto $|\downarrow\rangle$ yields a purely quadratic bosonic Hamiltonian,

$$H_{\text{np}} = \left(\omega - \frac{g^2}{\Delta}\right)a^\dagger a - \frac{g^2}{2\Delta}\left(aa + a^\dagger a^\dagger\right) - \Delta - \frac{g^2}{2\Delta}, \tag{9}$$

which a Bogoliubov transformation (Appendix B) diagonalizes, with $\xi = \omega - g^2/\Delta$ and $\kappa = -g^2/2\Delta$. Its GS is the squeezed vacuum $S_z|0\rangle \otimes |\downarrow\rangle$ with

$$r = -\frac{1}{4}\ln\left(1 - \frac{2g^2}{\omega\Delta}\right), \tag{10}$$

which diverges at g_c and is not real for $g > g_c$, so this treatment holds only in the normal phase. Undoing the SW rotation, the approximate GS of the QRM is

$$|\psi_{\text{probe}}\rangle = e^{-s_1}S_z|0\rangle \otimes |\downarrow\rangle. \tag{11}$$

QFI. Evaluating Eq. (2), the overlap term vanishes by symmetry and the derivative vector splits into orthogonal displacement and squeezing contributions, giving $F_Q = e^{-2r}/\Delta^2 + 2(\partial_g r)^2$, i.e.

$$F_Q(\varrho_g) = \frac{\sqrt{1 - \frac{2g^2}{\omega\Delta}}}{\Delta^2} + \frac{2g^2}{(\omega\Delta - 2g^2)^2}. \tag{12}$$

Close to the critical point $F_Q \approx \frac{1}{8}(g_c - g)^{-2}$, a critical exponent of -2. By the CRB, the coupling could in principle be estimated with vanishing error as $g \to g_c$. The same exponent is obtained in terms of γ.

B. Second-order transformation (SW2)

At third order, Eq. (A5) requires $[s_3, H_0] = \frac{g^3}{3\Delta^2}(a + a^\dagger)^3\sigma_x$, solved in the classical oscillator limit by

$$s_3 = -\frac{ig^3}{6\Delta^3}(a + a^\dagger)^3\sigma_y, \tag{13}$$

and Eq. (A6) gives the quartic Hamiltonian

$$H'_{\mathrm{SW2}} = \omega a^\dagger a + \Delta\sigma_z + \frac{g^2}{2\Delta}(a+a^\dagger)^2\sigma_z - \frac{7g^4}{24\Delta^3}(a+a^\dagger)^4\sigma_z. \tag{14}$$

A Bogoliubov transformation does not diagonalize a quartic Hamiltonian, so we expand $a \to \alpha + \delta a$ around the mean field and retain terms up to quadratic order in the fluctuations. In the normal phase the linear term fixes $\alpha = 0$ (the other root is unphysical), and the quadratic part has the same ξ and κ as in SW1, hence the same squeezing parameter. The approximate GS is $|\psi_{\mathrm{probe}}\rangle = e^{-s_1-s_3}S_z\,|0\rangle \otimes |{\downarrow}\rangle$. The derivative vector now has three contributions, with interference between the linear and cubic displacement terms (details in Ref. [11]), and the QFI is

$$F_Q(\varrho_g) = \frac{4g^2}{\Delta^4}\left(1 - \frac{2g^2}{\Delta\omega}\right) + \frac{1}{\Delta^2}\sqrt{1 - \frac{2g^2}{\omega\Delta}} + \frac{20g^4}{3\Delta^6}\left(1 - \frac{2g^2}{\Delta\omega}\right)^{3/2} + \frac{2g^2}{(\omega\Delta - 2g^2)^2}. \tag{15}$$

C. Comparison

Both results share the same diverging term. The regular parts, as functions of γ, are

$$\mathcal{I}^{\mathrm{SW1}} = \frac{\sqrt{1-2\gamma^2}}{\Delta^2}, \tag{16}$$

$$\mathcal{I}^{\mathrm{SW2}} = \frac{4\omega\gamma^2}{\Delta^3}(1-2\gamma^2) + \frac{\sqrt{1-2\gamma^2}}{\Delta^2} + \frac{20\omega^2\gamma^4}{3\Delta^4}(1-2\gamma^2)^{3/2}, \tag{17}$$

and are compared in Fig. 1. As $\omega \to 0$ they coincide, so in the strict classical oscillator limit the second-order correction brings no new information about the transition, although it improves the description of the QFI away from it.

V. NUMERICAL RESULTS

We now validate these predictions numerically, diagonalizing the full QRM in a truncated Fock space. Results are expressed in terms of γ at fixed $\Delta = 1$ and varying ω approaching the classical oscillator limit. Convergence is assessed through the fidelity $\mathcal{F} = |\langle\psi_N|\psi_{N-1}\rangle|^2$ between GSs of consecutive cutoffs (the padding procedure is documented in Ref. [11]).

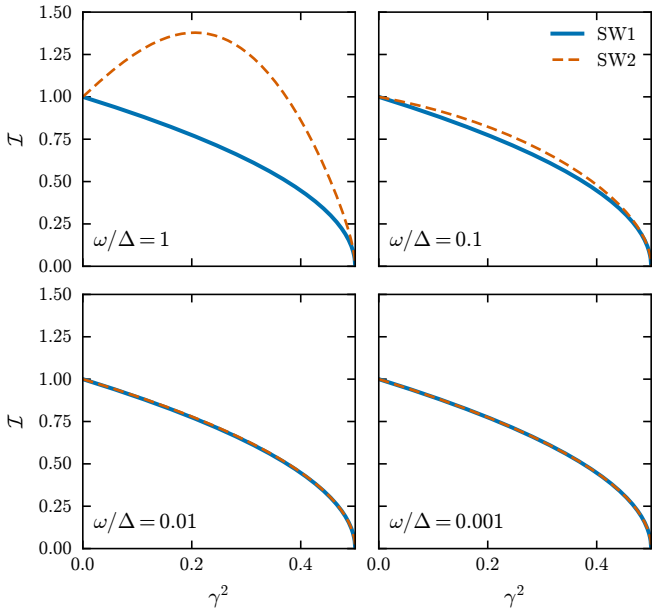


FIG. 1. Regular part of the QFI from the SW1 and SW2 Hamiltonians with $\Delta = 1$. The panels correspond to $\omega/\Delta = 1$, 0.1, 0.01 and 0.001.

A. Truncation and convergence

Figure 2 compares the two phases at $\gamma = \gamma_c \mp 0.05$. In the normal phase, a cutoff $N = 60$ suffices. In the superradiant phase the macroscopic photon occupation makes the standard Fock basis converge extremely slowly, so a different treatment is needed.

B. QFI in the normal phase

We compute the QFI from Eq. (2) with a finite-difference derivative whose step size is chosen to balance truncation and round-off errors (see Ref. [11] for the justification). Figure 3(a) shows the numerical QFI together with the analytic SW1 result, Eq. (12): the two agree over the whole normal phase, and both diverge on approaching the QCP. On a log-log scale [Fig. 3(b)] the QFI follows a straight line of slope -2, confirming the critical exponent [5]. Very close to γ_c the numerical curve departs from the divergence and saturates: the simulation uses a finite ω/Δ rather than the strict limit, so the system undergoes a smooth crossover and the non-analyticity is regularized. The heatmap in Fig. 4 shows that the enhancement persists well beyond the strict limit $\omega/\Delta \to 0$.

C. Superradiant phase: displaced and squeezed bases

The failure of SW1 beyond γ_c can be understood from Eq. (6). After projection onto the lower spin subspace, in

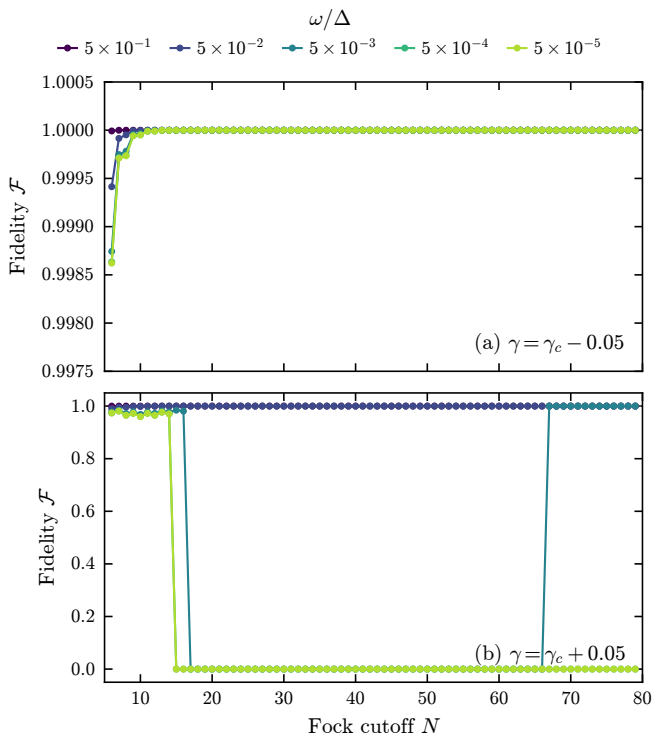


FIG. 2. Fidelity between consecutive truncations as a function of the cutoff for different ω/Δ with $\Delta = 1$: (a) normal phase and (b) superradiant phase.

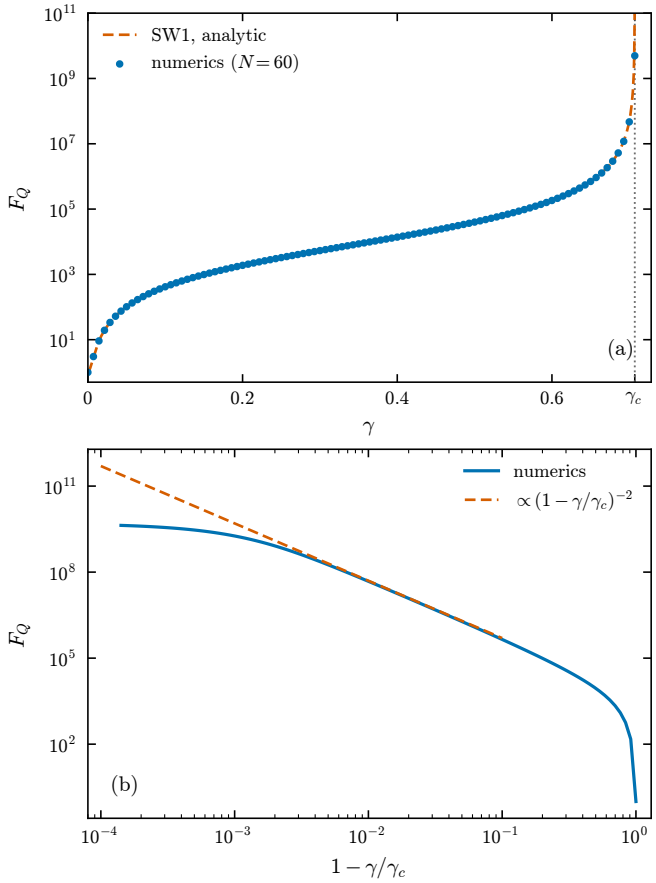


FIG. 3. QFI in the normal phase for $\omega/\Delta = 5 \cdot 10^{-5}$ ($N = 60$). (a) Numerical QFI of the full QRM (dots) and the analytic SW1 result of Eq. (12) (dashed line) as a function of γ. (b) Numerical QFI versus $1 - \gamma/\gamma_c$ on a log-log scale, compared with the SW1 asymptote of slope -2.

terms of quadratures,

$$H_b^{(\downarrow)} = \frac{\omega}{2}p^2 + \left(\frac{\omega}{2} - \frac{g^2}{\Delta}\right)x^2 + \text{const.}, \tag{18}$$

the coefficient of x^2 is proportional to $\gamma_c - \gamma$. For $\gamma > \gamma_c$ the potential is an inverted parabola, a Bogoliubov transformation is inapplicable, and the Fock basis centered at the origin is highly inefficient.

Displacement. We return to the full QRM and, following Ref. [5], write $H_{\text{QRM}} = \Delta[\frac{a^\dagger a}{\Delta/\omega} + \sigma_z + \gamma\frac{a+a^\dagger}{\sqrt{\Delta/\omega}}\sigma_x]$. For a separable variational GS $|\alpha\rangle \otimes |\psi\rangle$ with coherent state $\alpha = (x+iy)\sqrt{\Delta/\omega}$ and spin state $|\psi\rangle = \cos\theta\,|\uparrow\rangle + \sin\theta\,|\downarrow\rangle$, minimizing over θ and y leaves

$$f(x) = x^2 - \sqrt{1 + 4\gamma^2 x^2}, \tag{19}$$

shown in Fig. 5. Its minima are $x = 0$ in the normal phase and $x = \pm\sqrt{\gamma^2 - 1/(4\gamma^2)}$ in the superradiant phase, a spontaneous breaking of the parity symmetry. We therefore apply the displacement $D(\pm\alpha_s)$, with $\alpha_s = \sqrt{\gamma^2 - \frac{1}{4\gamma^2}}\sqrt{\Delta/\omega}$, to the Hamiltonian [5], using $D^\dagger(\alpha)aD(\alpha) = a + \alpha$ [17],

$$H_\pm = \omega a^\dagger a + \Delta\sigma_z \pm \omega\alpha_s(a + a^\dagger) + g\sigma_x(a + a^\dagger) \pm 2g\alpha_s\sigma_x + \omega\alpha_s^2. \tag{20}$$

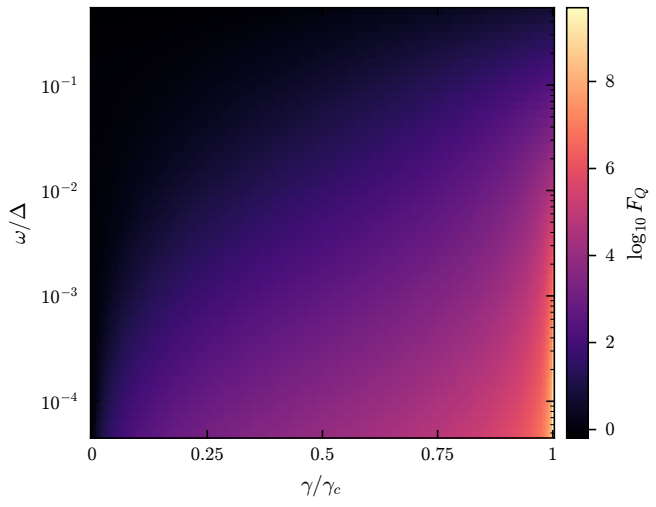


FIG. 4. QFI heatmap in the normal phase, showing $\log_{10} F_Q$ for 50 logarithmically spaced values of ω/Δ.

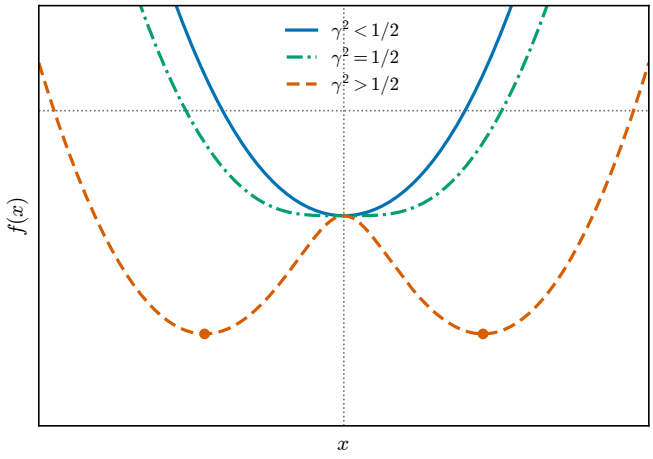


FIG. 5. Variational GS energy $f(x)$ below, at and above the critical point. In the superradiant phase the two minima (dots) signal the spontaneous symmetry breaking.

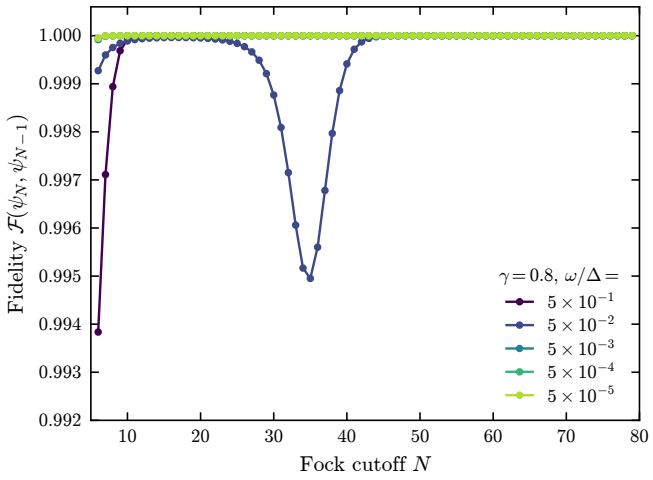


FIG. 6. Fidelity between consecutive bosonic truncations in the superradiant phase ($\gamma = 0.8$) for the displaced Hamiltonian.

This shifts the origin of phase space to the center of the state, so that the probability amplitude concentrates in the lowest Fock states. Figure 6 shows that, with this displacement, the fidelity is above 0.99 already at the smallest cutoffs shown, for all the values of ω/Δ considered.

Squeezing. To improve the convergence further, we rotate the spin basis, $\tau_i^{\pm} = e^{\mp i\theta\sigma_y}\sigma_i e^{\pm i\theta\sigma_y}$ with $\tan 2\theta = \mp 2g\alpha_s/\Delta$, so that $\Delta\sigma_z \pm 2g\alpha_s\sigma_x = \tilde{\Delta}\tau_z^{\pm}$ and $\tilde{\Delta} = \sqrt{\Delta^2 + (2g\alpha_s)^2}$. Then

$$H'_{\pm} = \omega a^\dagger a \pm \omega\alpha_s(a + a^\dagger) + \tilde{\Delta}\tau_z^{\pm} + g(a + a^\dagger)\left[\cos 2\theta\, \tau_x^{\pm} \mp \sin 2\theta\, \tau_z^{\pm}\right], \quad (21)$$

up to a constant. We apply a first-order SW transformation with $H_0 = \omega a^\dagger a + \tilde{\Delta}\tau_z^{\pm}$ and $H_1^{\pm}$ the remaining terms; the generator $s_1^{\pm} = aX - a^\dagger X^\dagger$ (with X acting on the spin) satisfying $[s_1^{\pm}, H_0] = -H_1^{\pm}$ is

$$s_1^{\pm} = (a - a^\dagger)\left[\mp\alpha_s + \frac{g\sin 2\theta}{\omega}\tau_z^{\pm} - \frac{\omega g\cos 2\theta}{\omega^2 - 4\tilde{\Delta}^2}\tau_x^{\pm}\right] + i(a + a^\dagger)\frac{2\tilde{\Delta}g\cos 2\theta}{\omega^2 - 4\tilde{\Delta}^2}\tau_y^{\pm}. \quad (22)$$

In the classical oscillator limit,

$$H_{\pm}^{\mathrm{SW1}} = \omega a^\dagger a + \tilde{\Delta}\tau_z^{\pm} \mp \frac{4g^2\alpha_s^2}{\tilde{\Delta}}\tau_z^{\pm} \pm \frac{g^2\Delta^2}{\tilde{\Delta}^3}(a + a^\dagger)^2\tau_z^{\pm}. \quad (23)$$

For H_+^{SW1} (the other sign is equivalent), the coefficient of $a^\dagger a$ in the negative τ_z^+ eigenspace, $\omega - 2g^2\Delta^2/\tilde{\Delta}^3$, is negative in the superradiant phase, so we project onto the positive eigenspace instead:

$$H_{\mathrm{eff}} = \left(\omega + \frac{2g^2\Delta^2}{\tilde{\Delta}^3}\right)a^\dagger a + \frac{g^2\Delta^2}{\tilde{\Delta}^3}\left(a^2 + (a^\dagger)^2\right). \quad (24)$$

Its Bogoliubov squeezing parameter is $r = \frac{1}{4}\ln\!\left(1 + \frac{4g^2\Delta^2}{\omega\tilde{\Delta}^3}\right)$, and the displaced Hamiltonian in the new bosonic operators $b, b^\dagger$ becomes

$$H_+ = \omega\cosh(2r)\, b^\dagger b - \frac{\omega}{2}\sinh(2r)\left(b^2 + (b^\dagger)^2\right) + e^{-r}(b + b^\dagger)\left[\omega\alpha_s + g\left(\cos 2\theta\, \tau_x^+ - \sin 2\theta\, \tau_z^+\right)\right] + \tilde{\Delta}\tau_z^+, \quad (25)$$

up to constants. The transformation relating H_+ to H_{QRM} is unitary, so H_+ has exactly the spectrum of the QRM; we verified numerically that its ground-state energy coincides with that of the exact diagonalization to machine precision. In this basis the fidelity is again above 0.99 for all cutoffs (Fig. 7). The displacement alone therefore already ensures convergence, and the squeezing mainly casts the Hamiltonian in the form that the Bogoliubov transformation of Appendix B diagonalizes.

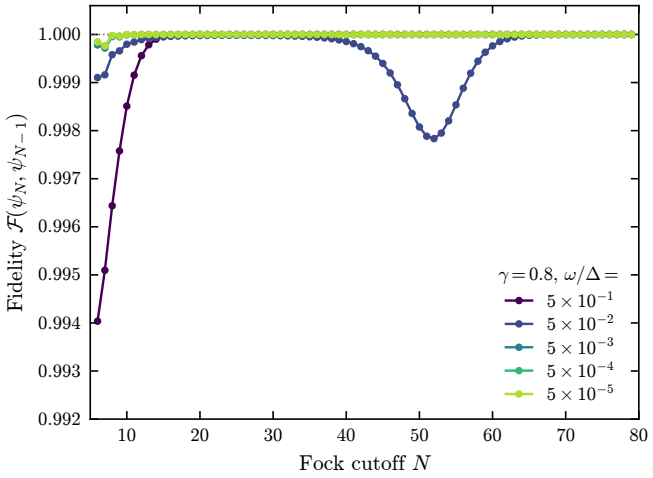


FIG. 7. Fidelity between consecutive bosonic cutoffs ($\gamma = 0.8$, $\Delta = 1$) for the displaced Hamiltonian written in the squeezed basis.

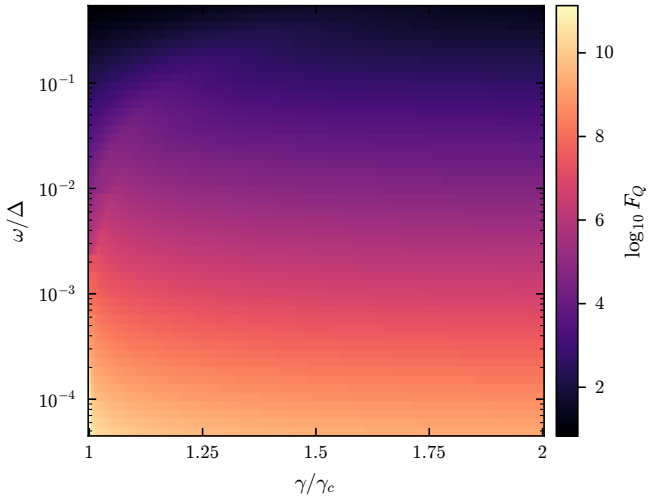


FIG. 8. QFI of the physical ground state in the superradiant phase for 50 logarithmically spaced values of ω/Δ. A thin strip near γ_c where the result depends on the Fock cutoff is interpolated.

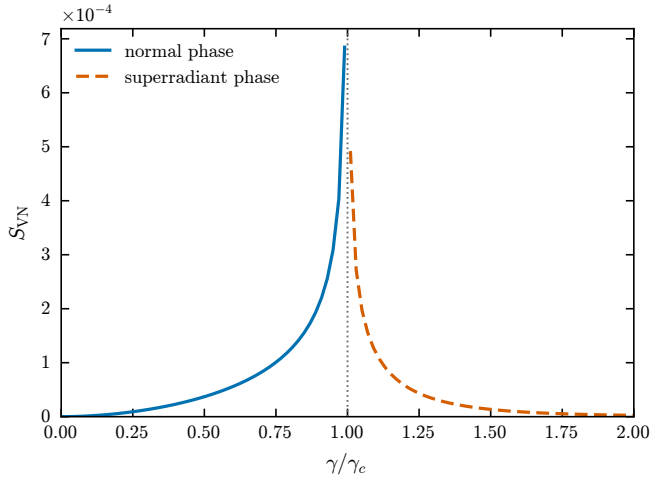


FIG. 9. Entanglement entropy in both phases for $\omega/\Delta = 5 \cdot 10^{-5}$ and $N = 50$.

QFI of the physical state. The displacement, the spin rotation and the squeezing all depend on g, so the physical ground state is $|\Psi(g)\rangle = D(\alpha_s)\, e^{i\theta\sigma_y}\, S_{-r}\, |\phi(g)\rangle$, with $|\phi\rangle$ the ground state of H_+ and S_{-r} the squeezing operator of Appendix B with $z = -r$. Differentiating, $\partial_g |\Psi\rangle = U(|\partial_g\phi\rangle + A\,|\phi\rangle)$ with $U = D\, e^{i\theta\sigma_y} S_{-r}$ and

$$A = (\partial_g \alpha_s)\, e^r (b^\dagger - b) + i(\partial_g\theta)\, \sigma_y + \tfrac{1}{2}(\partial_g r)\, (b^2 - b^{\dagger 2}), \quad (26)$$

so that Eq. (2) gives $F_Q = 4\big(\langle\chi|\chi\rangle - |\langle\phi|\chi\rangle|^2\big)$ with $|\chi\rangle = |\partial_g\phi\rangle + A\,|\phi\rangle$. Differentiating $|\phi\rangle$ alone, without the term A, underestimates the QFI by several orders of magnitude at small ω/Δ. We validated this expression against the QFI of the exact ground state of $H_{\rm QRM}$, computed in a very large Fock basis within its parity sector, for $\omega/\Delta \geq 0.01$: away from the critical point the two agree to better than 6×10^{-3} in $\log_{10} F_Q$.

Figure 8 shows the QFI of the superradiant phase for several ω/Δ bands. It is largest close to γ_c, decreases smoothly as γ increases, and grows as $\omega/\Delta \to 0$, reaching $\sim 10^{11}$ for $\omega/\Delta = 5 \cdot 10^{-5}$. The map is computed with three Fock cutoffs ($N = 60$, 80 and 100); in a thin strip close to γ_c (7% of the points), where the two symmetry-broken branches enter the truncated space, the results depend on the cutoff, and these points are interpolated. We find no enhanced-susceptibility regime other than the neighbourhood of the critical point.

D. Entanglement entropy

Having defined the numerical GSs in both phases with their optimal bases, we evaluate the von Neumann entanglement entropy between the atom and the field, $S_{\rm VN} = -\,\mathrm{Tr}(\rho_A \log_2 \rho_A)$, where ρ_A is the reduced density matrix of the atom [21]. The displacement, the spin rotation and the squeezing act locally on the field or on the atom, so they leave this entropy unchanged. Figure 9 shows that it is largest around γ_c and decreases on both sides, an independent signature of the transition [22]. In the normal phase $S_{\rm VN}$ grows monotonically up to 6.8×10^{-4} at the sampled point closest to the critical value ($\gamma/\gamma_c = 0.99$); in the superradiant phase it falls from 4.9×10^{-4} at $\gamma/\gamma_c = 1.01$ to nearly zero. The overall magnitude remains very small ($\lesssim 7 \times 10^{-4}$), confirming that the GS is well approximated by a separable mean-field ansatz in both phases, as expected in the classical oscillator limit. The sampling around γ_c is coarse, so the exact height and shape of the maximum are not resolved here.

VI. CONCLUSIONS

We have investigated critical quantum metrology with the quantum Rabi model, a finite-size system that exhibits critical behavior in the classical oscillator limit $\omega/\Delta \to 0$. First-order SW gives the critical coupling $g_c = \sqrt{\omega\Delta/2}$ and an analytic QFI diverging with exponent -2 in the normal phase. The second-order transformation refines the description away from the transition but converges to SW1 in the strict limit, so it does not alter the nature of the transition. In the superradiant phase, SW1 breaks down because of the inversion of the effective potential. Displacement, rotation and squeezing of the computational basis give a rapidly converging description of the superradiant phase, from which the QFI of the physical ground state follows once the coupling dependence of the transformation is accounted for, and the entanglement entropy, which peaks at the critical point, provides a complementary signature of the transition. These results support the finite-size QRM as a tractable platform for critical metrology, in which finite-component systems mimic thermodynamic-limit criticality.

This study addresses only the static precision limit. A complete assessment must include critical slowing down, since adiabatic GS preparation requires a time that diverges at the critical point. Future work should incorporate time-dependent dynamics to determine whether the precision gain justifies the cost of state preparation.

Appendix A: General Schrieffer-Wolff construction

Let $H = H_0 + \varepsilon H_1$, with H_0 block-diagonal (even) and H_1 block-antidiagonal (odd) [19]. We take $U = e^S$ with S anti-Hermitian and odd, so that

$$H_{\text{eff}} = e^S H e^{-S} = H + [S,H] + \frac{1}{2}[S,[S,H]] + \frac{1}{6}[S,[S,[S,H]]] + \dots \quad \text{(A1)}$$

Expanding $S = \sum_k \varepsilon^k s_k$ and using $[\mathcal{E},\mathcal{E}] = \mathcal{E}$, $[\mathcal{E},\mathcal{O}] = \mathcal{O}$, $[\mathcal{O},\mathcal{O}] = \mathcal{E}$ for even ($\mathcal{E}$) and odd ($\mathcal{O}$) operators, we require the odd part of each order in ε to vanish (ε is set to 1 at the end). The orders needed here are

$$\begin{aligned} H_{\text{eff}} = H_0 &+ \varepsilon\big(H_1 + [s_1,H_0]\big) \\ &+ \varepsilon^2\big([s_2,H_0] + [s_1,H_1] + \tfrac{1}{2}[s_1,[s_1,H_0]]\big) \\ &+ \varepsilon^3\Big([s_3,H_0] + [s_2,H_1] \\ &\quad + \tfrac{1}{2}\big([s_2,[s_1,H_0]] + [s_1,[s_2,H_0]] \\ &\qquad + [s_1,[s_1,H_1]]\big) \\ &\quad + \tfrac{1}{6}[s_1,[s_1,[s_1,H_0]]]\Big) + \mathcal{O}(\varepsilon^4). \end{aligned} \quad \text{(A2)}$$

First order. The odd term at order ε vanishes if

$$[s_1,H_0] = -H_1. \quad \text{(A3)}$$

At order ε^2 the only odd term is $[s_2,H_0]$, which forces $s_2 = 0$ (all even-indexed generators vanish [19]), and

$$H'_{\text{SW1}} = H_0 + \frac{\varepsilon^2}{2}[s_1,H_1]. \quad \text{(A4)}$$

Second order. With s_1 and $s_2 = 0$, the odd terms at order ε^3 vanish if

$$[s_3,H_0] = -\tfrac{1}{3}[s_1,[s_1,H_1]], \quad \text{(A5)}$$

and at order ε^4 one finds $s_4 = 0$, so that

$$\begin{aligned} H'_{\text{SW2}} &= H_0 + \tfrac{\varepsilon^2}{2}[s_1,H_1] \\ &\quad + \varepsilon^4\Big(\tfrac{1}{2}[s_3,H_1] + \tfrac{1}{8}[s_1,[s_1,[s_1,H_1]]]\Big) + \mathcal{O}(\varepsilon^5). \end{aligned} \quad \text{(A6)}$$

Appendix B: Squeezing operator, Bogoliubov transform and diagonalization

We define the squeezing operator for a single bosonic mode, $S_z = e^{(za^\dagger a^\dagger - z^* aa)/2}$ with $z = re^{i\theta}$, which is unitary and defines new bosonic operators (a Bogoliubov transform) [17]

$$\tilde{a} = S_z^\dagger a S_z = a\cosh r + a^\dagger e^{i\theta}\sinh r, \quad \text{(B1)}$$

$$\tilde{a}^\dagger = S_z^\dagger a^\dagger S_z = a^\dagger\cosh r + a e^{-i\theta}\sinh r. \quad \text{(B2)}$$

It diagonalizes Hamiltonians of the form $H = \xi a^\dagger a + \kappa a^\dagger a^\dagger + \kappa^* aa$. Applying the transformation to H and writing $\kappa = |\kappa|e^{i\phi}$ with the choice $\phi = \theta$,

$$\begin{aligned} S_z^\dagger H S_z &= (\xi\cosh 2r + |\kappa|\sinh 2r)\, a^\dagger a \\ &\quad + \Big(\tfrac{\xi}{2}\sinh 2r + |\kappa|\cosh 2r\Big)\,(e^{i\phi}a^\dagger a^\dagger + e^{-i\phi}aa) \\ &\quad + |\kappa|\sinh 2r + \xi\sinh^2 r. \end{aligned} \quad \text{(B3)}$$

The terms in $a^\dagger a^\dagger$ and aa vanish for

$$\tanh(2r) = -\frac{2|\kappa|}{\xi}, \quad r = \frac{1}{2}\arctan\left(\frac{2|\kappa|}{\xi}\right) \quad \text{or} \quad r = -\frac{1}{4}\ln\left(\frac{\xi - 2|\kappa|}{\xi + 2|\kappa|}\right), \quad \text{(B4)}$$

and the Hamiltonian becomes

$$H' = \tilde{\xi}a^\dagger a + \frac{\tilde{\xi} - \xi}{2}, \qquad \tilde{\xi} = \sqrt{\xi^2 - 4|\kappa|^2}, \quad \text{(B5)}$$

a harmonic oscillator with GS $|0\rangle$ and energy $E_0 = (\tilde{\xi} - \xi)/2$. Hence $S_z^\dagger H S_z |0\rangle = E_0 |0\rangle$, and multiplying by S_z shows that the exact GS of H is $S_z|0\rangle$.

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